

Plasma concentrations of homocysteine and other aminothiols are related to food intake in healthy human subjects.

We investigated total, free and protein-bound plasma homocysteine, cysteine and cysteinylglycine in 13 subjects aged 24-29 y after a breakfast at 0900 h containing 15-18 g of protein and a dinner at 1500 h containing approximately 50 g of protein. Twelve subjects had normal fasting homocysteine (mean $\pm$ SD, 7.6 $\pm$ 1.1 μ mol/L) and methionine concentrations (22.7 $\pm$ 3.5 μ mol/L) and were included in the statistical analyses. Breakfast caused a small but significant increase in plasma methionine (22.2 $\pm$ 20.6%) and a brief, nonsignificant increase followed by a significant decline in free homocysteine. However, changes in total and bound homocysteine were small. After dinner, there was a marked increase in plasma methionine by 16.7 $\pm$ 8.9 μ mol/L (87.9 $\pm$ 49%), which was associated with a rapid and marked increase in free homocysteine (33.7 $\pm$ 19.6%, 4 h after dinner) and a moderate and slow increase in total (13.5 $\pm$ 7.5%, 8 h) and protein-bound (12.6 $\pm$ 9.4%, 8 h) homocysteine. After both meals, cysteine and cysteinylglycine concentrations seemed related to changes in homocysteine, because there were parallel fluctuations in the free:bound ratios of all three thiols. Dietary changes in plasma homocysteine will probably not affect the evaluation of vitamin deficiency states associated with moderate to severe hyperhomocysteinemia but may be of concern in the risk assessment of cardiovascular disease in patients with mild hyperhomocysteinemia. Synchronous fluctuations in the free:bound ratio of the plasma aminothiol compounds indicate that biological effects of homocysteine may be difficult to separate from effects due to associated changes in other aminothiol compounds.